

1. Name and ages

```
students={ "Ram":20,"Sneha":19,"Rahul":20,"Riya":18,"Sreena":20}  
print(students["Sneha"])
```

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2.Add and update key value pairs

```
my_dic={}  
my_dic["a"]=23  
my_dic["b"]=2  
my_dic["c"]=12  
my_dic["b"]=30  
print(my_dic)
```

{'a': 23, 'b': 30, 'c': 12}

3. Check if key exists

```
students={ "Ram":20,"Sneha":19,"Rahul":20,"Riya":18,"Sreena":20}  
search_key="Ram"  
if search_key in students:  
    print(students[search_key])  
else:  
    print("Keys not found")
```

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4. Dictionary with 10 items

```
d={  
    "a":5,  
    "b":4,  
    "c":3,  
    "d":9,  
    "e":12,  
}  
print("Number of key-value pairs:",len(d))
```

Number of key-value pairs: 5

5. Print all keys,values,pairs

```
data={"Name":"Ram","age":20}  
print(data)
```

```
print(data["Name"])
print(data.get('age'))
```

```
{'Name': 'Ram', 'age': 20}
Ram
20
```

6.List

```
keys=["name","age","city"]
values=["Sneha",20,"Kolkata"]
data=dict(zip(keys,values))
print(data)
```

```
{'name': 'Sneha', 'age': 20, 'city': 'Kolkata'}
```

7. Nested Dictionary

```
employees = {
    "employee1":{
        "name":"Sneha",
        "age":21,
        "salary":50000
    },
    "employee2":{
        "name":"Raj",
        "age":20,
        "salary":60000
    },
    "employee3":{
        "name":"Ratul",
        "age":22,
        "salary":70000
    }
}
print("Salary of second employee:",
employees["employee2"]["salary"])
```

```
Salary of second employee: 60000
```

8. Count the Frequency

```
text="LOVE"
freq={}
for ch in text:
    freq[ch]=freq.get(ch,0)+1
print("Character Frequency:",freq)
```

```
Character Frequency: {'L': 1, 'O': 1, 'V': 1, 'E': 1}
```

9.DICTIONARY COMPREHENSION FOR SQUARES FROM 1 TO 10

```
squares={x:x**2 for x in range(1,11)}  
print(squares)
```

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64, 9: 81, 10: 100}
```

10.Merge two Dictionaries

```
dict1 = {"a": 10, "b": 20}  
dict2 = {"b": 30, "c": 40}
```

```
merged = {**dict1, **dict2} # Keeps value from dict2 on duplicate keys  
print(merged)
```

```
{'a': 10, 'b': 30, 'c': 40}
```

11.Safely Remove Key

```
d = {"a": 1, "b": 2, "c": 3}  
key_to_remove = "b"
```

```
removed_val = d.pop(key_to_remove, None)  
print("Updated Dictionary:", d)
```

```
Updated Dictionary: {'a': 1, 'c': 3}
```

12.Highest Score Student

```
scores = {"Ram": 85, "Sneha": 95, "Rahul": 78, "Riya": 92}  
top_student = max(scores, key=scores.get)  
print("Top Student:", top_student, "with score:", scores[top_student])
```

```
Top Student: Sneha with score: 95
```

13.Swap Keys and Values

```
original = {"a": 1, "b": 2, "c": 1}  
swapped = {}
```

```
for key, val in original.items():  
    if val not in swapped:  
        swapped[val] = [key]  
    else:  
        swapped[val].append(key)
```

```
print(swapped)
```

```
{1: ['a', 'c'], 2: ['b']}
```

14.Filter by Threshold

```
data = {"a": 15, "b": 5, "c": 25, "d": 8}
threshold = 10

filtered = {k: v for k, v in data.items() if v > threshold}
print(filtered)

{'a': 15, 'c': 25}
```

15.Group Words by First Letter

```
words = ["apple", "banana", "avocado", "cherry", "blueberry"]
grouped = {}

for word in words:
    first_letter = word[0].lower()
    grouped.setdefault(first_letter, []).append(word)

print(grouped)

{'a': ['apple', 'avocado'], 'b': ['banana', 'blueberry'], 'c': ['cherry']}
```

16.Safe Access with get()

```
user_info = {"name": "Rahul", "age": 22}

city = user_info.get("city", "Not Specified")
print("City:", city)

City: Not Specified
```

17.Word Frequency in Sentence

```
import string

sentence = "Hello world! Hello python, world."
clean_sentence = sentence.translate(str.maketrans("", "", string.punctuation))
words = clean_sentence.lower().split()

word_freq = {}
for word in words:
    word_freq[word] = word_freq.get(word, 0) + 1

print(word_freq)

{'hello': 2, 'world': 2, 'python': 1}
```

18.Dictionary with List Values

```
categories = {"fruits": [], "vegetables": []}

categories["fruits"].append("Apple")
categories["fruits"].append("Banana")
```

```
categories["vegetables"].append("Carrot")
```

```
print(categories)
```

```
{'fruits': ['Apple', 'Banana'], 'vegetables': ['Carrot']}
```

19.Sort Dictionary by Values

```
d = {"a": 3, "b": 1, "c": 2}
```

```
asc_sorted = dict(sorted(d.items(), key=lambda item: item[1]))
```

```
desc_sorted = dict(sorted(d.items(), key=lambda item: item[1], reverse=True))
```

```
print("Ascending:", asc_sorted)
```

```
print("Descending:", desc_sorted)
```

```
Ascending: {'b': 1, 'c': 2, 'a': 3}
```

```
Descending: {'a': 3, 'c': 2, 'b': 1}
```

20.Remove Duplicates Preserving Order

```
lst = [1, 3, 2, 3, 1, 4, 5, 2]
```

```
unique_lst = list(dict.fromkeys(lst))
```

```
print(unique_lst)
```

```
[1, 3, 2, 4, 5]
```